Preliminary

replaced by revised version

PROJECT 7000

April 9, 1959

FILE MEMO

SUBJECT: Letter from Dr. R. B. Lazarus of LASL
Dated March 27, 1959

Dr. Lazarus' letter lists a number of questions, remarks and requests concerning STRETCH. They may be classified into two catagories: (1) suggestions for hardware changes and new instructions, and (2) requests for information as to how the computer operates in certain special cases.

Some of the questions have been answered in telephone conversations with Dr. Lazarus, but the complete list is given here for the record.

A. Concerning Equipment:
  I. Space Available
  The number of operation codes not being used are, according to class:

1.	Floating Point	3	
2.	Miscellaneous	3	
3.	Direct and Immediate Index	0	
4.	Count & Br. and Indicator Br.	1	(BB combination)
5.	Unconditional Branch	1	(enter stream)
6.	VFL Connect	2	
7.	Input/Output	6	
8.	Transmit	1+5/16	
9.	Branch	1	(enter stream)
10.	Miscellaneous	8	
11.	Overflow Indicator	32	

Many of the above have modifier bits belonging to the particular class of instruction so they actually represent many more cases. Care must be taken in how the above op-codes are considered "free" however, because the instruction decoding is done in several partial steps.

Q: How does a program entered by basic linkage determine the enable-disable states of machine?

Concerning the CONVERT instructions and Decimal arithmetic, the CONVERTS were added because they resulted in an appreciable simplification in hardware and made a powerful tool available to the programmer. Having only one CONVERT instruction which operated on the accumulator would require the programmer to write a number of extra load and store instructions to preserve the accumulator contents, etc.

Decimal VFL arithmetic enables the system to handle a whole class of problems directly without subroutines and is considered to be an important advantage in the STRETCH system.

II. New Orders Desired

In general the STRETCH design effort is well beyond the stage where new instruction sequences can be added. The truth tables of these items are complete and the logical design is almost finished.

Nevertheless, the suggested orders have been examined carefully by all the engineers involved and appraised for the hardware changes which would be required. The following statements are direct quotations from their comments:

(same as cnct to Mem) no add ?

1. Swap with Accumulator
   This is not a swap instruction. It would be done with a VFL operation which would be highly unique but similar in some respects to a Connect to Memory. It would be very awkward and slow. ~~[illegible]~~ can be done in 3 half wds.

M → A
A → M
A → A
look into

2. Store Floating Point Zero
   This appears to be similar to the Store zero instruction but would require a modification of data paths. A transmit instruction does it in a full word.

(Good)

3. Set Exponent; Set Exponent Immediate
   It would have been easy but the accumulator is built. ← Paul will look into

√Q: Acc → Acc . sq rt should leave in acc, since this

— possibility of making acc — except & fast

Q√ st. sq rt: speed — hardware not ready.

underflow: "oring" bit into high order position,

4. **Reflect Accumulator**

This is within the potential of the VFL unit, however, the paths involved are completed. A further comment is that it would be one bit at a time (byte size = 1) and the accumulator would be addressed as an operand (internal memory) making the instruction quite slow.

can program comp (~~invert~~) subtract, compare two acc's

5. **Assorted Shift Orders**

The machine is not set up for circular shifts. This would be a real change in the design of our floating point unit. Also it would be complicated by amount of shift and a left or right option. (See the comments under 7 below). — ring shift

store
store
load

6. **Set Indicator One**

The VFL unit must always be used for this and, therefore, branch on bit or indicator branch is equivalent in result and speed.

decision Dec'57 invert — lose protection of read-only indicators
- can do with BB or connect

7. **Extract**

Masking, extracting and shifting are another logical way of doing the same tasks accomplished by our VFL instructions. The VFL philosophy was chosen for STRETCH over two years ago as being a more general and efficient method. Suggestions that we abandom VFL in favor of extracts or put in both systems at this late date are equivalent to saying we should start all over again on the logical design.

bit 55 mean "forward or back"

8. **Generalized Swaps and Transmits**

This would get extremely awkward where the transmission involved the memory addresses of the index storage since the X register is used for all index words. Proper design of the transmit instruction would call for a slow down on all transmits due to the special actions that need be taken under these special conditions. — also give up half word TMT

any Δ, no [illegible] [illegible],

What about — TMT & SWAP neg. might be easy (Bloch will check) need bit 55

9. **Connect to Sign Register**

This instruction was in the design at an earlier date but was removed because of the hardware complexity. It can be done now in two instructions using the accumulator.

(needed switching matrix.

✓ Q: why none on FP compare? - -
not on unnorm.

1. Load Ind. n times

2. Load to E time

10. Locate List Element

The idea of counting for each load indirect would require a revision of the instruction sequences for Load Indirect. The effect of "load indirect to the n th level" and "Load indirect until you find element E" can be done with relatively simple programs.

can be programmed several ways

[to go thru list & search until find E or end of list]

III. Miscellaneous Comments and Questions

1. Q: Does a VFL Store or Connect to Memory require a complete fetch of the old data word, or is the information merged at the memory unit? Does a two bit store to an address with bit address 63 require two full fetch bus times and two full store bus times?

A: No merging of information at the memory unit takes place. The example given requires two fetches and two stores. The main effect of this is on bus traffic. The machine normally overlaps fetches and stores with other operations.

2. Q: What is the timing for a Swap of n words between a and a + 1? In particular, when step two is reached, does Stretch have to re-fetch from a + 1 the word it just stored there? We would like this to be made reasonably fast, if possible

A: Timing is 4.8 n microsecond. Yes, the word is refetched. Progress of Swap is as follows: $Z_L + 1 = Z_R$

Address	a	a + 1	a + 2	a + 3	
Original	b	c	d	e	
1st	c	b	d	e	
2nd	c	d	b	e	etc.

3. Q: If an ordinary floating point number is added to a zero with large negative exponent, is the Preparatory Shift Greater than 48 indicator turned on?

A: Yes. The preparatory shift $>$ 48 indicator is based on a difference is exponents with no exceptions.

4. Q: IBM has still never replied to our request to give us an example justifying the existence of the Set and Wait instruction.

A: In practice Set and Wait is used with its own location in the address position in order to be sure the wrong interrupt doesn't send the computer off. Branch enable to itself gives the same result but the STOP lite is not turned on.

should be in manual

5. Q: What is the present indicator set? Is there a Noisy Mode indicator which can be masked on or off? We feel that there must be, for safety. We request very strongly that all unused indicators (in particular, the "Harvest" indicators) be wired up as programmers' indicators.

decided to include

A: The noisy mode indicator (#63) is masked off. The Harvest indicators simply lack Harvest inputs. The new Chap. 3 gives the present indicator set.

very [illegible] point.

6. Q: When a memory operand is at an address less than 16, is there a gain in speed? We would like things to be arranged so that one can use the miscellaneous registers for data and get fast access.

A: No. The concept that transistorized registers will yield fast access to data in a random fashion was a strong fallacy in the early planning of the STRETCH computer. More advanced planning has revealed that the orientation of the machine to receive data from core memory and fundamental logical contradictions make access from transistorized storage even slower than 2.0 usec. core storage in many cases.

machine is designed to speed up reg mem. ops. lookahead & multiplexing

7. Q: What happens if there are two op-code 8's in succession? If this gives Invalid Operation, then we are wasting full word op-codes.

A: Does give invalid operation. Necessary to avoid ambiguous cases.

→ ✓ Q: why 1 millisec on [illegible]? 1 millisec to 2 tied to clock

✓ Q: TMT does it interrupt on clock? yes

✓ Q: BB does it store back? Yes, always store back

✓ Q: how fast is swap in ½ μs mem? 12 cycles.

8. Q: How much time is spent for the two boundary register tests? Is this slowing down the machine during otherwise fast loops? Where is this comparison made?

A: Boundary register tests do not cause a slow down due to some extra equipment in the look-ahead unit which stores the result of the test and takes appropriate action. The comparison is actually made in the memory bus unit. This time is overlapped with Memory box selection, etc.

9. Q: What is the present status of the floating point Add to Magnitude and Augment orders? Have you made the changes analagous to those made in the integer set? We feel that this should be done.

A: New chapter 4 describes present status.

10. Q: What are the reasons for not having fast one-word fixups? Can we have them if we prohibit the programmer from changing data that is used in the next few succeeding instructions?

A: Fast one-word fixups implies the existance of a special look-ahead register to hold them. This would make multiple interrupts logically impossible. Programming restrictions such as that due to suggested are contrary to fundamental STRETCH philosophy.

11. Q: What happens if a Rename instruction has an effective address identical to the address in the current X0 Refill field? It is essential that the store effectively precede the fetch, as it says in the manual.

Store precede the fetch

rewrite A: ~~The fetch preceeds the store.~~ This is more compatible with the general way in which storing is done in the machine, making the instruction simpler and faster.

12. Q: What happens if a VFL Immediate instruction, such as Load, specifies a Field Length greater than 24 bits?

A: Fills in zeros in low order.

– Set up high neg in either case of { mp cand. either first
dividend either
add no change

13. Q: What are the properties of Floating Point Zeros in connection with Exponent Underflow? If there is a mode of machine operation in which any result having negative exponent spill is replaced quickly and automatically by a number with exponent all ones and mantissa zero, then we have no further worries. If there is not such a mode, then the fixup must be via interrupt. In the days of the fast one-word fixup, it was checked that everything was all right. If fix-ups always run the look-ahead dry, then things would still be all right if, and only if, a negative exponent spill indication is not given by the product of an ordinary number with negative exponent and a zero with exponent all ones. Similarly, no indication should be given when dividing a zero with exponent all ones by an ordinary number with positive exponent.

A: A Floating Point zero (zero fraction) is independent of exponent underflow in the circuit logic. There is no machine mode for generating the negative exponent and zero fraction. The fixup is assumed to be by interrupt. The underflow indication is set very late in the instruction sequence, too late to effect the starting of another sequence. Changing the logic would slow down all floating point instructions. Multiply and divide are not special cases as far as exponent spill is concerned. The same tests are made on all floating point instructions.

14. Q: Why are accumulator bits 60-63 cleared in some single precision floating point operations? This can be irritating in some cases, and we should like to know whether it is really necessary, please explain why it is desirable, or else change things so as to leave 60-127 of the accumulator alone.

A: These bits are used in "double" orders but not in single. The mixing of double and single results implies "coding trick" type of result which we have intentionally tried to avoid. From an engineering point of view, the handling of bits 60 to 63 can no longer be open to review. Too much of the designed portions of the machine have been affected.

14 – [illegible] [illegible] – would have to read out & back – (no final ans.,

15. Q: Is there any reason why bit 55 is not doing more work in the Transmit orders?

A: Has been changed to full word transmits and swaps.

16. Q: We are delighted to see a new full word internal register and two new indicators. The register could be a handy temporary storage spot, presumably fast, especially if it were equipped with its own Load from and Store to memory. It would also be tempting to make it an indicator register.

A: The N register is an address in 2 usec. core storage and load N is of the same order of speed and arrangement as load cumulative multiplicand. Due to the desire to eliminate certain minor inconsistancies both instructions are quite slow and awkward to perform. Since it is not a transistorized register, it could not serve as an indicator register very effectively. Adding new transistorized registers is impossible at this late date.

17. Q: What happens on interrupt if the Interruption Address is less than 32?

A: The Indicator A0 is turned on. If next address is also below 32, is stuck in loop.

18. Q: What happens if the interrupt table is in a protected area? Is the Instruction Fetch indicator set?

A: Gets table value, no indicators turned on.

19. Q: Is there any special protection for the boob who has an interrupt table of all binary zeros?

A: No. He is stuck in a loop.

20. Q: What happens when compound orders modify themselves? For example: at address a there is a VFL Store with Progressive Indexing; the value field of index I has a and the address field of the VFL Store has β ; is the final value field of index I 2a or a + β ?

A: $a + \beta'$.

Q: Another example: in a there is a STICI to a + 1/2; in a + 1/2 there is a half-word Branch to β ; do we branch to β all right? If so, suppose β is not protected, but a + 1 is; do we get any false indicators?

A: If there is a STICI in a, then A + 1/2 contains the branch address of that instruction. If this address is B, then the machine will branch to B. If it is desired to store the instruction counter in A + 1/2 there will be no problem, since this is done by means of a store to a after the instruction counter has been placed in the contents of a.

21. Q: What happens if you put a pair of half-word orders into the interrupt table? Are they both performed, or only the first? If both, what happens if the first causes an interrupt?

A: Only the first one is executed under any circumstances.

22. Q: What happens if there is a data store to, or fetch from, a.63, say, of a field longer than one bit, and a is not protected but a + 1 is? Or a full word instruction fetched from a + 1/2?

A: Whole field is protected if any part of it falls outside the boundary. Instruction is "no-op'ed".

EIJ:HGK:jcj

E. I. Jordan

✓Q: TMT type which drifts into protected area? — it stops at boundary — doesn't start again.

✓Q: TMT in & out of ind regs — when some ~~some~~ inds are set by TMT not To Mem, sets inds then wipes out by store.

H. G. Kolsky

— fetch comes before anything is changed.
— store comes after? (lookahead)